SWAGAT NEUPANE

Fremont, CA | neupaneswagat@gmail.com
github.com/Swagaat | swagaat.github.io

SUMMARY

Computer science student graduating Dec 2026 with experience in Python software development and applied machine learning. Owns RestroPilot's ML workflow, connecting data preparation and model evaluation with tested inference interfaces and React application integration.

EDUCATION

San Francisco Bay University | Fremont, CA

Bachelor of Science in Computer Science | Expected Dec 2026

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL (coursework), HTML, CSS

Software: React, Git, GitHub, pytest, Vitest, Zod, JSON contracts

ML and Data: pandas, NumPy, scikit-learn, joblib, feature engineering, MAE, WAPE

PROJECTS

RestroPilot | ML Developer | Team capstone | Offline research prototype

Python, pandas, scikit-learn, joblib, pytest, React, TypeScript | Source

- Built a CSV reconciliation pipeline preserving 194,001 order lines across two restaurants, with stable source keys, exact decimal price handling, quality flags, and checksum manifests.
- Engineered calendar features, 1/7/14/28-day lags, and shifted rolling means; trained a scikit-learn Random Forest pipeline using chronological splits and training-only item selection.
- Reduced historical test MAE from 5.569 to 4.172 units versus a weekly baseline on 239 scored item-days out of 366 calendar item-days using rolling one-day evaluation.
- Designed checksum-bound saved-model inference that validates artifact compatibility and 28 consecutive days of history, rejecting stale inputs and unvalidated items without retraining.
- Integrated results into React through a Python JSON exporter and Zod-validated fetch layer, with charts, CSV export, and unavailable-data states; tested leakage, artifact integrity, and interface errors.

Personal Portfolio and Interactive Web Projects

HTML, CSS, JavaScript, SVG | Source

- Built a responsive multi-page portfolio with shared navigation, project case studies, and a keyboard-accessible JavaScript calculator.
- Implemented category filters and an interactive SVG place explorer for a Nepal showcase, with accessible selection states and content available without JavaScript.

Bay Area Environmental Data Analysis | Academic prototype

Python, pandas, Matplotlib, Seaborn, scikit-learn | Source

- Explored pollution and asthma indicators through missing-value handling, county comparisons, and correlation plots; implemented exploratory Random Forest and Gradient Boosting regression experiments.

LEADERSHIP AND ACTIVITIES

House Captain and Class Leader, Uniglobe Higher Secondary School: led 10 students and helped organize a chess competition and academic quiz.